

Project Title
Transformer-Based Aspect-Based Sentiment Analysis
of Product Reviews

Project Report submitted
For the partial fulfilment of grade for
the subject

Applied Machine Learning for Text Analysis (24ALT3101)

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ABSTRACT

Product reviews contain valuable information about customer opinions, experiences, and satisfaction regarding different features of a product. Traditional sentiment analysis methods generally assign a single sentiment label to an entire review, which may not accurately capture opinions expressed toward individual product features. This project proposes a **Transformer-Based Aspect-Based Sentiment Analysis (ABSA)** system for identifying specific aspects mentioned in product reviews and determining the sentiment associated with each aspect. The proposed system utilizes pre-trained Transformer models such as **BERT, RoBERTa, or DistilBERT** to capture contextual relationships within review text and perform fine-grained sentiment analysis. The system involves data preprocessing, text tokenization, contextual representation, aspect extraction, and aspect-level sentiment classification. For example, a review such as *“The camera quality is excellent, but the battery life is poor”* can be analyzed as **Camera Quality → Positive** and **Battery Life → Negative**. The model is evaluated using performance measures such as accuracy, precision, recall, and F1-score. The resulting aspect-sentiment pairs can be further aggregated to identify commonly appreciated features, customer complaints, and areas requiring product improvement. The proposed system aims to provide more detailed and actionable insights from product reviews compared with traditional document-level sentiment analysis.

PROBLEM STATEMENT

Online product reviews contain detailed opinions about multiple features of a product, but traditional sentiment analysis usually assigns a single sentiment label to the entire review. This approach fails to identify the different sentiments expressed toward individual aspects of a product. For example, a customer may state that *“The camera quality is excellent, but the battery life is poor.”* A conventional sentiment classifier may classify the complete review as positive or negative, without distinguishing the sentiment associated with the camera and battery. Therefore, there is a need for an intelligent system that can identify specific product aspects from customer reviews and determine the sentiment expressed toward each aspect. The proposed project addresses this problem by developing a Transformer-based Aspect-Based Sentiment Analysis system that uses contextual language representations to perform accurate

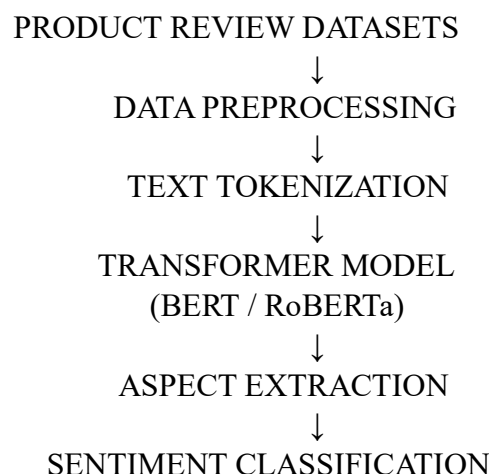
aspect extraction and aspect-level sentiment classification. The system can help transform unstructured product reviews into meaningful insights that can support customers, businesses, and product developers in understanding product strengths and weaknesses.

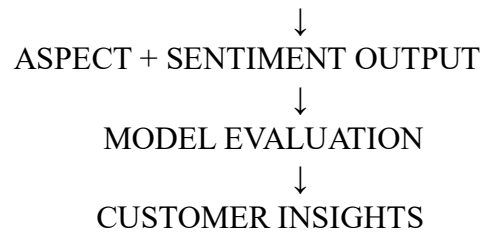
OBJECTIVES

The main objectives of the proposed project are:

1. **To preprocess product review data** by removing noise and preparing textual data for Transformer-based analysis.
2. **To identify and extract product aspects** mentioned in customer reviews, such as camera, battery, display, price, performance, and design.
3. **To determine the sentiment associated with each identified aspect** and classify it as positive, negative, neutral, or other relevant sentiment categories.
4. **To utilize pre-trained Transformer models** such as BERT, RoBERTa, or DistilBERT for understanding the contextual meaning of product reviews.
5. **To generate aspect-sentiment pairs** from reviews to provide fine-grained information about customer opinions.
6. **To evaluate the performance of the proposed model** using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
7. **To aggregate aspect-level sentiment results** to identify frequently praised features and commonly reported problems.
8. **To provide meaningful customer and product insights** that can assist businesses and product developers in improving product quality and customer satisfaction.

Flowchart (Project Structure):





CONCLUSION

The proposed **Transformer-Based Aspect-Based Sentiment Analysis of Product Reviews** provides a fine-grained approach to understanding customer opinions by identifying individual product aspects and determining the sentiment associated with each aspect. Unlike traditional sentiment analysis, which assigns a single sentiment to an entire review, the proposed system can distinguish multiple opinions within the same review. Transformer models such as BERT, RoBERTa, or DistilBERT can effectively capture the contextual relationships between words, enabling accurate aspect extraction and sentiment classification.

The system can generate meaningful aspect-sentiment pairs such as **Camera Quality → Positive** and **Battery Life → Negative**, allowing customer feedback to be analyzed in greater detail. The performance of the model can be evaluated using accuracy, precision, recall, and F1-score. The resulting insights can help identify frequently appreciated product features, common customer complaints, and areas that require improvement. Overall, the proposed system demonstrates how Transformer-based Natural Language Processing can convert large volumes of unstructured product reviews into useful and actionable information for customers, businesses, and product developers.

SOFTWARE REQUIREMENTS

Software Requirements

The following software and technologies are required for implementing the proposed **Transformer-Based Aspect-Based Sentiment Analysis of Product Reviews** system:

Category	Software / Technology
Operating System	Windows 10/11, Linux, or macOS
Programming Language	Python 3.10 or above
Development Environment	Jupyter Notebook / Google Colab / VS Code
NLP Framework	Hugging Face Transformers
Deep Learning Framework	PyTorch or TensorFlow

Category	Software / Technology
Data Processing	Pandas, NumPy
Natural Language Processing	NLTK / spaCy
Machine Learning	Scikit-learn
Data Visualization	Matplotlib, Seaborn
Dataset	SemEval-2014 ABSA Dataset, Amazon Product Reviews
Pre-trained Models	BERT / RoBERTa / DistilBERT
Model Evaluation	Accuracy, Precision, Recall, F1-score, Confusion Matrix
Version Control	Git and GitHub
Optional Interface	Streamlit or Flask

Minimum Software Setup

- **Python:** 3.10+
- **Jupyter Notebook or Google Colab:** For model development and experimentation
- **PyTorch:** For training and running Transformer models
- **Transformers:** For accessing pre-trained BERT/RoBERTa/DistilBERT models
- **Pandas & NumPy:** For dataset processing
- **Scikit-learn:** For evaluation and supporting machine-learning operations
- **Matplotlib & Seaborn:** For visualization of results
- **NLTK/spaCy:** For text preprocessing and linguistic processing
- **Git/GitHub:** For project version control and code management